Mitochondrial DNA Depletion Syndrome 12B (MTDPS12B): A Comprehensive Disease Characterization

Disease: Mitochondrial DNA Depletion Syndrome 12B (cardiomyopathic type), autosomal recessive **Causal gene: SLC25A4** (encoding ANT1/AAC1, the heart- and skeletal-muscle-specific mitochondrial ADP/ATP translocase) **MONDO:** MONDO:0014175 · **OMIM:** 615418 · **Synonyms:** MTDPS12B; mitochondrial DNA depletion syndrome 12 (cardiomyopathic type), AR **Inheritance:** Autosomal recessive · **Category:** Mendelian

Summary

Mitochondrial DNA Depletion Syndrome 12B (MTDPS12B; MONDO:0014175; OMIM 615418) is an **ultra-rare autosomal-recessive mitochondrial disorder caused by biallelic loss-of-function variants in SLC25A4**, the gene encoding **adenine nucleotide translocator 1 (ANT1, also called AAC1)** — the heart- and skeletal-muscle-predominant carrier that exchanges mitochondrial matrix ATP for cytosolic ADP across the inner mitochondrial membrane. It is critical to note that *SLC25A4* is the correct causal gene (not a similarly numbered mitochondrial depletion syndrome from a different locus): the disease name is a serial number, and the identifiers above (SLC25A4; OMIM 615418) govern. Recessive null alleles abolish ADP/ATP exchange, producing a bioenergetic ("energy crisis") defect accompanied by secondary mitochondrial DNA instability (multiple mtDNA deletions and reduced mtDNA copy number) that is largely restricted to oxidative, high-energy-demand tissues — the heart and skeletal muscle.

Clinically, MTDPS12B presents in infancy or childhood with **hypertrophic cardiomyopathy**, **mitochondrial myopathy** with exercise intolerance, and **lactic acidosis**; one complete-null patient additionally had **congenital cataracts**. Muscle biopsy shows the hallmarks of mitochondrial myopathy — ragged-red fibers, proliferation of structurally abnormal mitochondria, increased mitochondrial mass, and multiple mtDNA deletions. Notably, patients

lack the chronic progressive external ophthalmoplegia (CPEO) that characterizes the *dominant* SLC25A4 disorders, defining a clear recessive-versus-dominant genotype–phenotype dichotomy.

The recessive cardiomyopathic form (MTDPS12B) is mechanistically and clinically distinct from the dominant SLC25A4 diseases: heterozygous dominant-negative missense alleles cause adult-onset autosomal-dominant PEO with multiple mtDNA deletions, while recurrent *de novo* dominant alleles (p.Arg80His, p.Arg235Gly) cause a severe neonatal depletion disorder (MTDPS12A, OMIM 617184). The *Ant1*-knockout mouse faithfully recapitulates the human cardiomyopathic/myopathic phenotype — including lactic acidosis, mitochondrial hyperproliferation, OXPHOS induction, elevated ROS, and excessive mtDNA damage — providing a validated model system. There is currently **no disease-specific or curative therapy**; management is supportive.

Section 1 — Disease Information

Overview. MTDPS12B is a Mendelian mitochondrial disease in which inherited biallelic loss of the heart/skeletal-muscle ADP/ATP translocase cripples the export of mitochondrially produced ATP to the cytosol. Energy-demanding striated muscle (cardiac and skeletal) is preferentially affected, yielding an infantile/childhood-onset syndrome of cardiomyopathy and myopathy with lactic acidosis. The disorder sits within the broader family of mitochondrial DNA maintenance disorders because affected tissue accumulates **multiple mtDNA deletions** and shows **reduced mtDNA copy number** as a secondary consequence of the primary transport defect.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0014175
OMIM	615418
Gene (HGNC)	SLC25A4 (HGNC:10990)
Protein (UniProt)	P12235 (ADT1_HUMAN / ANT1)
Synonyms	MTDPS12B; mitochondrial DNA depletion syndrome 12 (cardiomyopathic type), AR; ANT1 deficiency, recessive

Data source type. The evidence base is derived from **individual patient case reports** (aggregated into disease-level resources such as OMIM) combined with **model-organism and in-vitro functional studies**, rather than from large EHR cohorts. Because the disorder is ultra-rare, the human literature comprises a small number of index families.

Section 2 — Etiology

Primary cause (genetic). MTDPS12B is caused exclusively by **biallelic (autosomal recessive) loss-of-function of SLC25A4/ANT1** (Finding F001). Two mechanisms of complete functional loss are documented: (1) a homozygous missense allele that abolishes transport activity (c.368C>A, p.Ala123Asp), and (2) a homozygous splice-donor abolition (c.111+1G>A) that eliminates ANT1 expression entirely. In the first recessive report, "*No ATP uptake was measured in proteoliposomes reconstituted with protein extracts from the patient's muscle*" [PMID: 16155110], and the equivalent mutation in the yeast ortholog AAC2 abolished transport — establishing true loss of function. In the complete-null case, "*Complete loss of expression of the ANT1 gene causes a clinical syndrome mainly characterised by cardiomyopathy and myopathy*" [PMID: 22187496].

Genetic risk factors. The only established risk factor is **inheritance of two pathogenic SLC25A4 alleles** (homozygous or compound heterozygous). Consanguinity increases risk of homozygosity for recessive alleles, consistent with the reported homozygous cases. No susceptibility loci or common modifier variants have been established for this ultra-rare disorder.

Environmental risk factors. None established. The disease is monogenic; onset and expression are driven by the genotype rather than environmental exposure. Age and the high intrinsic metabolic demand of striated muscle act as biological context rather than modifiable exposures.

Protective factors. No genetic or environmental protective factors are established in humans. At the mechanistic level, expression of the paralogous translocases **ANT2 (SLC25A5)** and **ANT3 (SLC25A6)** could in principle buffer ANT1 loss; in the mouse, higher *Ant2* expression in extraocular muscle may spare ocular motility (see Sections 6 and 15). However, in the human complete-null patient, "*No compensating increase in the transcript level of the ANT3 gene … was observed*" [PMID: 22187496], so paralog compensation is tissue-dependent and incomplete.

Gene–environment interactions. None documented. This is a highly penetrant monogenic disorder with no reported GxE modulation.

Section 3 — Phenotypes

The phenotypic spectrum (Finding F004) is dominated by striated-muscle and metabolic manifestations. Across the two index recessive cases:

Phenotype	Type	Suggested HPO	Onset	Severity / course	Frequency
Hypertrophic cardiomyopathy	Clinical sign / cardiac	HP:0001639	Infantile–childhood	Severe, progressive	Core feature (both cases)
Mitochondrial myopathy / muscle weakness	Clinical sign	HP:0003198 / HP:0003737	Infantile–childhood	Mild→moderate, progressive	Core feature (both cases)
Exercise intolerance	Symptom	HP:0003546	Childhood	Variable	Reported
Lactic acidosis	Laboratory abnormality	HP:0003128 (elevated lactate)	Infantile–childhood	Moderate–severe	Core feature (both cases)
Ragged-red fibers	Laboratory/ histopathology	HP:0003200	—	—	Present on biopsy
Multiple mtDNA deletions	Laboratory/ molecular	HP:0003689 (mtDNA deletion)	—	—	Present in muscle
Congenital cataracts	Clinical sign / ocular	HP:0000519	Congenital	—	Complete-null case
Absence of ophthalmoplegia	Distinguishing feature	(cf. HP:0000602)	—	—	Notably absent (vs dominant PEO)

Supporting quotes: a "*sporadic patient who presented with hypertrophic cardiomyopathy, mild myopathy with exercise intolerance and lactic acidosis but no ophthalmoplegia*" [PMID: 16155110]; and "*an autosomal recessive mitochondrial disorder characterised by congenital cataracts, hypertrophic cardiomyopathy, myopathy and lactic acidosis*" [PMID: 22187496].

Quality-of-life impact. Although no formal EQ-5D/SF-36 data exist for this ultra-rare disease, the combination of hypertrophic cardiomyopathy (risk of heart failure/arrhythmia), progressive myopathy with exercise intolerance, and metabolic (lactic) acidosis implies substantial functional limitation from early life, with cardiac disease being the principal determinant of morbidity and prognosis.

Section 4 — Genetic / Molecular Information

Causal gene. SLC25A4 (solute carrier family 25 member 4), chromosome 4q35.1, encoding ANT1/AAC1 (UniProt P12235), a ~298-residue inner-mitochondrial-membrane carrier of the SLC25 mitochondrial carrier family with the characteristic three tandem ~100-residue repeats and six transmembrane helices.

Pathogenic variants (recessive, MTDPS12B).

Variant (cDNA)	Protein	Type	Functional consequence	Classification	Source
c.368C>A	p.Ala123Asp	Missense	Abolished ADP/ATP transport (no ATP uptake in proteoliposomes); yeast AAC2 equivalent abolished transport; absent in 500 controls	Pathogenic	[PMID: 16155110]
c.111+1G>A	— (splice donor loss)	Splice-site	Complete loss of ANT1 expression; transcript undetectable in muscle and fibroblasts	Pathogenic	[PMID: 22187496]

Both recessive alleles produce **complete loss of function**. Origin is **germline**; no somatic contribution is relevant. Allele frequencies in gnomAD are vanishingly rare/absent, consistent with the ultra-rare recessive disease.

Contrast — dominant SLC25A4 alleles (NOT MTDPS12B). Heterozygous missense alleles cause adult-onset autosomal-dominant PEO (adPEO / PEOA2) with multiple mtDNA deletions, acting through a dominant-negative mechanism; recurrent *de novo* dominant alleles **c.239G>A**

(p.Arg80His) and **c.703C>G (p.Arg235Gly)** cause severe neonatal MTDPS12A (OMIM 617184) with marked loss of mtDNA copy number (Finding F002) [PMID: 27693233]. Recent structural work maps the "lethal" mutations near the primary substrate (S1) site and "milder" PEO mutations near a secondary (S2) site, correlating mutation location with severity [PMID: 42308345].

Functional consequence. Loss of function (recessive MTDPS12B). The dominant disorders by contrast behave as dominant-negative/gain-of-toxicity.

Modifier genes. The paralogs **ANT2 (SLC25A5)** and **ANT3 (SLC25A6)** are candidate functional modifiers via dosage compensation, though compensation is tissue-restricted and was absent for ANT3 in affected human muscle [PMID: 22187496].

Epigenetic / chromosomal abnormalities. None reported; MTDPS12B is a point-mutation/splice disorder, not a structural/cytogenetic one.

Section 5 — Environmental Information

No environmental factors, lifestyle factors, or infectious agents are implicated in MTDPS12B. It is a purely genetic, monogenic recessive disorder. (Environmental mitochondrial toxins are relevant to acquired mitochondrial dysfunction generally but are not causes of this inherited syndrome.)

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function mutation in SLC25A4** (missense p.Ala123Asp or splice-null c.111+1G>A) → **leads to** absent or non-functional ANT1/AAC1 protein in heart and skeletal muscle.
2. Loss of functional ANT1 → **results in** abolished ADP/ATP exchange across the inner mitochondrial membrane (demonstrated: no ATP uptake in patient-muscle proteoliposomes) → matrix ATP cannot be exported and cytosolic ADP cannot be imported.

3. Failure of adenine-nucleotide exchange → **leads to** a cellular **bioenergetic/energy crisis** in high-demand oxidative tissue (cytosolic ATP shortfall; impaired re-supply of ADP substrate to ATP synthase).
4. (Branch A — bioenergetic) Energy crisis → **results in** lactic acidosis and impaired contractile function of cardiac and skeletal muscle.
5. (Branch B — mtDNA instability, partly inferred) Impaired adenine-nucleotide/matrix nucleotide homeostasis → **probably affects mitochondrial DNA maintenance → leads to** multiple mtDNA deletions and reduced mtDNA copy number → loss of respiratory-chain complexes containing mtDNA-encoded subunits → further aggravates the energy crisis (feed-forward loop).
6. Chronic bioenergetic stress → **results in** compensatory mitochondrial hyperproliferation (ragged-red fibers, increased mitochondrial mass), OXPHOS enzyme induction, and increased reactive oxygen species (ROS) → **leads to** additional oxidative mtDNA damage (demonstrated in the mouse).
7. Cumulative cardiac and skeletal-muscle energetic failure → **results in** the clinical manifestation: **hypertrophic cardiomyopathy, mitochondrial myopathy with exercise intolerance, lactic acidosis** (± congenital cataracts).

> *Branch B (step 5) is the mechanism stated as "probably" in the literature* — the mtDNA-maintenance link is inferred/associative rather than fully demonstrated: "*This highly reduced capacity for transport probably affects mitochondrial DNA maintenance and in turn respiration, causing a severe energy crisis*" [PMID: 27693233].

Detail by category

- **Molecular pathways / biochemistry.** The core defect is in the adenine-nucleotide translocation step of **oxidative phosphorylation (OXPHOS)**. ANT1 "*exports mitochondrial ATP to the cytosol*" in exchange for ADP [PMID: 18439414]; loss blocks the ATP/ADP shuttle that couples matrix ATP synthesis to cytosolic energy demand. Downstream compensation engages the mitochondrial biogenesis program — in the *Ant1*−/− mouse there is "*coordinate upregulation of Pgc-1α, Nrf-1, Tfam, myogenin and antioxidant genes*" [PMID: 18439414].
- **Cellular processes.** Mitochondrial hyperproliferation/biogenesis; oxidative stress; secondary apoptosis signaling (ANT1 is a component of the mitochondrial permeability transition pore). GO terms: GO:0140021 (mitochondrial ADP transmembrane transport),

GO:0006839 (mitochondrial transport), GO:0006119 (oxidative phosphorylation), GO:0007005 (mitochondrion organization), GO:0000002 (mitochondrial genome maintenance), GO:0006979 (response to oxidative stress).

- **Protein dysfunction.** Loss of transport function — either by abolished catalytic activity (p.Ala123Asp) or absent protein (splice-null). Structural mapping indicates substrate-binding/architecture disruption underlies severity gradients across the allelic series [PMID: 42308345].
- **Metabolic changes.** Impaired ATP export → lactic acidosis (anaerobic compensation), respiratory-chain deficiency for mtDNA-encoded subunits.
- **Tissue damage mechanisms.** Oxidative stress (increased ROS), excessive mtDNA damage, energy-deprivation-driven cardiomyocyte hypertrophy and myofiber degeneration.
- **Immune involvement.** None; this is a non-inflammatory metabolic myopathy/cardiomyopathy.

Upstream vs downstream. Upstream = the SLC25A4 mutation and loss of ADP/ATP exchange. Downstream = energy crisis, mtDNA instability, ROS, mitochondrial proliferation, and ultimately cardiomyopathy/myopathy.

Cell types involved (CL): cardiomyocyte (CL:0000746), skeletal muscle fiber (CL:0008002 / CL:0000188), lens fiber cell (CL:0011004, for cataract).

Section 7 — Anatomical Structures Affected

- **Primary organs:** heart (UBERON:0000948) and skeletal muscle (UBERON:0001134). Cardiovascular and musculoskeletal systems are primarily affected.
- **Secondary involvement:** ocular lens (UBERON:0000965) — congenital cataracts in the complete-null case; systemic metabolic effect via lactic acidosis.
- **Tissue/cell level:** striated (cardiac and skeletal) muscle tissue; cardiomyocytes (CL:0000746) and skeletal myofibers (CL:0000188); lens fiber cells.
- **Subcellular level: mitochondrion** (GO:0005739), specifically the **inner mitochondrial membrane** (GO:0005743) where ANT1 resides; the mitochondrial matrix (nucleotide pools) and the mitochondrial genome (mtDNA maintenance) are secondarily affected.

- **Localization / lateralization:** bilateral/systemic involvement of striated muscle; cardiomyopathy affects the whole myocardium; cataracts are bilateral. The disease is not lateralized.

The tissue restriction is explained by **isoform-specific expression**: "*The ANT1 isoform predominates in heart and muscle while ANT2 is systemic*" [PMID: 35714845] (Finding F006).

Section 8 — Temporal Development

- **Onset:** congenital to early childhood (infantile/childhood-onset). Congenital cataracts in one case indicate prenatal/neonatal onset of at least one feature; recessive cardiomyopathic SLC25A4 disease is described as **childhood-onset** [PMID: 27693233].
- **Onset pattern:** chronic/insidious with progressive deterioration; metabolic crises (lactic acidosis) can be episodic.
- **Progression:** progressive myopathy and cardiomyopathy; muscle shows accumulating mtDNA deletions and mitochondrial proliferation over time. Progression rate is variable but the cardiomyopathy is the life-limiting component.
- **Disease course:** chronic, lifelong, progressive.
- **Critical periods:** infancy/early childhood, when cardiac decompensation is most threatening.
- **Remission:** no spontaneous remission; no disease-modifying treatment exists to induce remission.

Section 9 — Inheritance and Population

- **Inheritance: autosomal recessive** (Finding F001). The heterozygous carrier mother in the complete-null family had "*normal clinical, biochemical and pathological features*," confirming no carrier phenotype [PMID: 22187496].
- **Penetrance / expressivity:** biallelic LOF appears fully penetrant for the striated-muscle/metabolic phenotype; expressivity varies (e.g., cataracts present in one case, not emphasized in the other).

- **Epidemiology: ultra-rare.** Only a small number of recessive index cases are reported worldwide; precise prevalence/incidence figures are not established (too few cases). No reliable per-100,000 estimate exists.
- **Carrier frequency / founder effects:** not established; pathogenic recessive alleles are essentially absent from population databases (gnomAD).
- **Consanguinity:** relevant — homozygous recessive cases are consistent with consanguineous or founder backgrounds, though not detailed for every case.
- **Sex ratio:** no sex predilection expected (autosomal); too few cases to quantify.
- **Genetic anticipation / germline mosaicism:** not applicable/not reported (no repeat expansion).

Section 10 — Diagnostics

Clinical/laboratory tests. - **Blood lactate:** elevated (lactic acidosis) — HP:0003128; a key biochemical clue. - **Cardiac evaluation:** echocardiography/MRI demonstrating hypertrophic cardiomyopathy; ECG. - **Muscle biopsy histopathology:** ragged-red fibers; COX/SDH histochemistry; proliferation of structurally abnormal mitochondria; increased mitochondrial mass [PMID: 16155110; 22187496]. - **Respiratory-chain enzymology:** deficiency of complexes containing mtDNA-encoded subunits [PMID: 27693233]. - **Functional transport assay (research):** proteoliposome ATP-uptake assay from patient muscle demonstrated absent transport [PMID: 16155110].

Molecular/mtDNA testing. - **mtDNA analysis** (long-range PCR / Southern blot / sequencing): reveals **multiple mtDNA deletions** and **reduced mtDNA copy number** — the molecular signature of an mtDNA-maintenance disorder. - **Nuclear gene testing:** single-gene **SLC25A4** sequencing, or (preferably) a **mitochondrial/nuclear mtDNA-maintenance gene panel**, **whole-exome (WES)**, or **whole-genome sequencing (WGS)** to identify biallelic SLC25A4 variants. Because the phenotype overlaps many mitochondrial disorders, broad panel/exome testing is the practical first-line genetic approach.

Clinical criteria / differential diagnosis. No disease-specific criteria. Differential diagnosis includes other causes of **infantile hypertrophic cardiomyopathy + lactic acidosis**: other mtDNA-maintenance/depletion syndromes (e.g., other SLC25A4 disorders, POLG, MPV17, TK2,

AGK/Sengers syndrome — note Sengers also features congenital cataracts + HCM), respiratory-chain complex deficiencies, and fatty-acid-oxidation defects (e.g., VLCADD). The recessive SLC25A4 form is distinguished from dominant SLC25A4 disease by the **absence of ophthalmoplegia** and the childhood cardiomyopathic presentation.

Screening. Carrier/cascade testing within affected families once the pathogenic alleles are known; prenatal/preimplantation genetic testing is feasible. No population newborn screening targets this ultra-rare disorder.

Section 11 — Outcome / Prognosis

No formal survival statistics exist for this ultra-rare disease. Prognosis is driven primarily by the **hypertrophic cardiomyopathy**, which carries risk of heart failure and arrhythmia from early life; the myopathy contributes chronic disability and exercise intolerance, and recurrent lactic acidosis adds metabolic morbidity. The disease is chronic, progressive, and lifelong. There is no established disease-specific mortality figure; cardiac involvement is the principal determinant of outcome. **Prognostic factors** likely include severity and progression of cardiomyopathy and the completeness of ANT1 loss. No validated prognostic biomarkers beyond serial cardiac imaging and lactate monitoring are established.

Section 12 — Treatment

No disease-specific or curative therapy exists. Management is **supportive and symptomatic**:

- **Cardiac care:** standard heart-failure management for hypertrophic/dilated cardiomyopathy; monitoring for arrhythmia; consideration of advanced cardiac support/transplantation in end-stage cardiac failure (as applied generically in mitochondrial cardiomyopathies).
- **Metabolic support:** management of lactic acidosis; "mitochondrial cocktail" supplements (e.g., coenzyme Q10, L-carnitine, riboflavin) are used empirically across mitochondrial myopathies/cardiomyopathies, though no controlled evidence establishes efficacy specifically in MTDPS12B. (NCIT candidate terms: Coenzyme Q10 therapy, Levocarnitine, Riboflavin.)

- **Supportive/rehabilitative:** physiotherapy for myopathy; activity modification for exercise intolerance; cataract surgery if visually significant.
- **Avoidance** of mitochondrial toxins where possible.

Advanced therapeutics. No approved gene, cell, RNA, or targeted therapy. Because the recessive form is a **loss-of-function** disorder, **gene replacement/augmentation** of SLC25A4 is a conceptually attractive but currently experimental strategy. No disease-specific clinical trials (NCT identifiers) are established.

Pharmacogenomics / personalized medicine. Not applicable beyond confirming biallelic LOF genotype to guide family counseling.

Section 13 — Prevention

- **Primary prevention:** not possible (inherited monogenic disorder). Prevention centers on **reproductive genetic counseling**.
- **Genetic counseling / reproductive options:** autosomal-recessive recurrence risk of 25% for couples who are both carriers; **carrier screening, prenatal diagnosis, and preimplantation genetic testing (PGT)** are available once the familial SLC25A4 variants are identified.
- **Secondary/tertiary prevention:** early cardiac surveillance and heart-failure management to prevent decompensation; metabolic management to limit acidotic crises; cataract monitoring.
- **Immunization / public-health / environmental interventions:** not applicable.

Section 14 — Other Species / Natural Disease

- **Orthologous genes:** mouse *Slc25a4* (*Ant1*); yeast *AAC2* (the functional ortholog used to validate the human p.Ala123Asp mutation) [PMID: 16155110].
- **Model/natural disease:** the *Ant1*−/− **mouse** (NCBI Taxon 10090) is an engineered model (see Section 15) rather than a naturally occurring disease. No well-characterized naturally occurring companion-animal or wildlife form of recessive ANT1 deficiency is established in the reviewed literature.

- **Evolutionary conservation:** ANT/AAC function (ADP/ATP exchange across the inner mitochondrial membrane) is deeply conserved from yeast to human, underpinning the utility of the yeast AAC2 and mouse *Ant1* systems for modeling human disease.
- **Zoonotic potential:** not applicable (non-infectious genetic disease).

Section 15 — Model Organisms

Mouse (*Ant1*−/− knockout). This is the principal model and faithfully recapitulates the human cardiomyopathic/myopathic phenotype (Finding F003). The knockout "*results in lactic acidosis and mitochondrial cardiomyopathy and myopathy, the latter involving hyper-proliferation of mitochondria, induction of oxidative phosphorylation (OXPHOS) enzymes, increased reactive oxygen species (ROS), and excessive mtDNA damage*" [PMID: 18439414], with coordinate upregulation of the mitochondrial-biogenesis and antioxidant programs. *Ant1*−/− skeletal muscle shows ragged-red-fiber-like mitochondrial myopathy [PMID: 16303948].

Phenotype recapitulation and limitations. The mouse reproduces the cardiomyopathy, myopathy, ragged-red fibers, mitochondrial proliferation, ROS, and mtDNA damage of the human recessive disease. A key **species difference** is ocular: human *dominant* ANT1 disease (CPEO) preferentially affects extraocular muscle, yet *Ant1*−/− mice have "*no measurable ocular motor abnormalities*," apparently because "*EOMs of wild-type mice exhibited higher levels of Ant2 mRNA … which may compensate for the Ant1 loss*" [PMID: 16303948] — directly illustrating the **ANT2 paralog-compensation** principle (Finding F006). Retinal function is also largely preserved in *Ant1*−/− mice, with supernormal ERG b-waves but normal morphology, suggesting ANT1 is not essential for retinal ATP transport [PMID: 20671283].

Other models. The **yeast *AAC2*** system provided orthogonal validation that the human p.Ala123Asp mutation abolishes transport [PMID: 16155110]. *Ant1* heterozygous conditional-knockout mice have additionally been used to probe neuropsychiatric links (bipolar-disorder-associated serotonergic changes) tied to ANT1 haploinsufficiency [PMID: 29892051], although this is tangential to the recessive cardiomyopathic disease.

Research applications. These models allow dissection of the bioenergetic defect, mtDNA-maintenance failure, ROS biology, compensatory biogenesis, and tissue-specific paralog compensation — and provide a platform for testing gene-replacement and metabolic therapies.

Key Findings (expanded)

F001 — Biallelic loss-of-function of SLC25A4/ANT1 is the cause

MTDPS12B is caused by **autosomal-recessive, complete loss of function** of ANT1. The first recessive report described a homozygous c.368C>A (p.Ala123Asp) variant absent from 500 controls, with proteoliposome assays showing no ATP uptake and the yeast AAC2 equivalent abolishing transport [PMID: 16155110]. A later complete-null case documented homozygous c.111+1G>A splice-donor abolition with ANT1 transcript undetectable in muscle and fibroblasts [PMID: 22187496]. Together these establish LOF as the disease mechanism.

F002 — Recessive and dominant SLC25A4 alleles cause distinct diseases

A clear allelic dichotomy exists: "*Several heterozygous SLC25A4 mutations cause adult-onset autosomal-dominant progressive external ophthalmoplegia associated with multiple mitochondrial DNA deletions, whereas recessive SLC25A4 mutations cause childhood-onset mitochondrial myopathy and cardiomyopathy*" [PMID: 27693233]. Recurrent *de novo* dominant alleles (p.Arg80His, p.Arg235Gly) cause severe neonatal MTDPS12A (OMIM 617184) with marked mtDNA-copy-number loss. This is central to correct diagnosis and classification.

F003 — The Ant1−/− mouse recapitulates the human phenotype

The knockout mouse develops lactic acidosis, mitochondrial cardiomyopathy and myopathy, mitochondrial hyperproliferation, OXPHOS induction, increased ROS, excessive mtDNA damage, and coordinate upregulation of Pgc-1α/Nrf-1/Tfam/myogenin/antioxidant genes [PMID: 18439414], with ragged-red-fiber-like myopathy [PMID: 16303948] — validating it as a disease model.

F004 — Clinical phenotype: HCM, myopathy, lactic acidosis, ± cataracts

Core features are hypertrophic cardiomyopathy, mitochondrial myopathy with exercise intolerance, and lactic acidosis, **without ophthalmoplegia** [PMID: 16155110]; the complete-null case added **congenital cataracts** [PMID: 22187496]. Biopsies show ragged-red fibers, abnormal mitochondrial proliferation, increased mitochondrial mass, and multiple mtDNA deletions. The heterozygous carrier mother was unaffected, confirming recessivity.

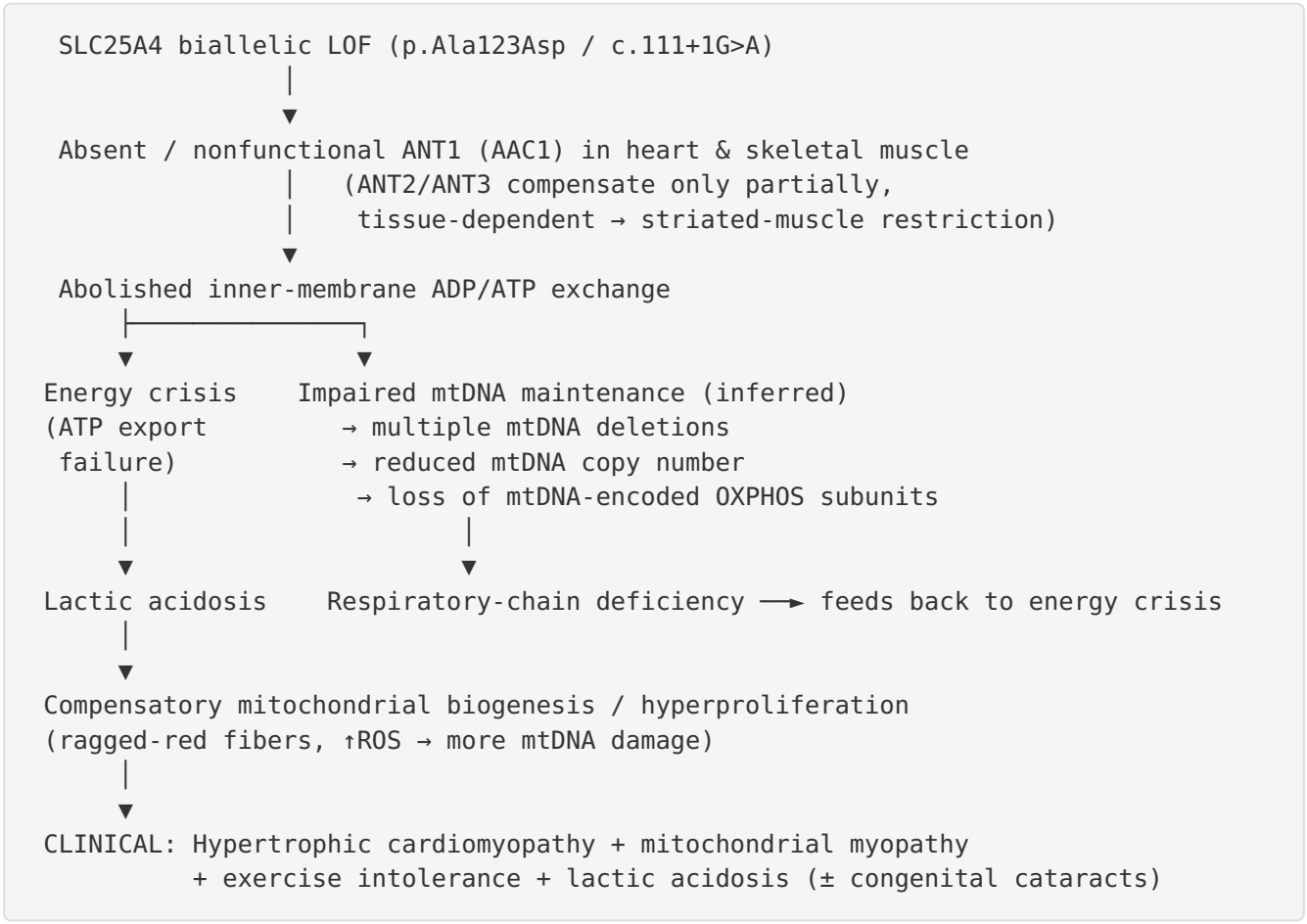
F005 — Mechanism: transport loss → energy crisis + secondary mtDNA instability

ANT1 normally "*exports mitochondrial ATP to the cytosol*" [PMID: 18439414]; recessive LOF abolishes ADP/ATP exchange. The reduced transport "*probably affects mitochondrial DNA maintenance and in turn respiration, causing a severe energy crisis*" [PMID: 27693233], with loss of mtDNA-encoded respiratory-chain complexes and reduced mtDNA copy number. Structural work maps disease mutations onto the carrier's substrate-binding architecture, correlating location with severity [PMID: 42308345].

F006 — Tissue specificity from isoform expression; partial paralog compensation

"*The ANT1 isoform predominates in heart and muscle while ANT2 is systemic*" [PMID: 35714845], explaining the striated-muscle-restricted disease. Compensation is incomplete and tissue-dependent: no ANT3 upregulation occurred in affected human muscle [PMID: 22187496], whereas higher *Ant2* in mouse extraocular muscle may spare ocular motility [PMID: 16303948].

Mechanistic Model



Upstream: SLC25A4 mutation → transport loss. **Downstream:** energy crisis, mtDNA instability, ROS, mitochondrial proliferation → cardiomyopathy/myopathy.

Evidence Base

PMID	Title (abbreviated)	Role in this report
PMID: 16155110	Complete LOF of heart/muscle ANT associated with myopathy & cardiomyopathy	Establishes recessive LOF (p.Ala123Asp), transport-null proteoliposomes, yeast validation; core phenotype without ophthalmoplegia
PMID: 22187496	Complete loss of ANT1 expression causing cardiomyopathy & myopathy	Splice-null complete-null case; adds congenital cataracts; no ANT3 compensation; recessive (unaffected carrier mother)
PMID: 27693233	Recurrent de novo dominant SLC25A4 mutations (MTDPS12A)	Distinguishes recessive vs dominant disease; energy-crisis/mtDNA-maintenance mechanism
PMID: 18439414	MITOCHIP in Ant1-KO muscle	Mouse phenotype + biogenesis/antioxidant program; defines ANT1 ATP-export function
PMID: 16303948	Ant1-KO: CPEO pathology, normal ocular motility	Ragged-red-fiber myopathy; ANT2 paralog compensation in EOM
PMID: 35714845	ANT transporters in myogenesis	Isoform expression (ANT1 heart/muscle; ANT2 systemic) explaining tissue specificity
PMID: 42308345	Structural basis of AAC substrate transport & disease heterogeneity	Maps severity to substrate-binding sites (S1 lethal, S2 milder)
PMID: 20671283	Retinal function in Ant1-deficient mice	Retina largely spared; ANT1 not essential for retinal ATP transport
PMID: 29892051	Ant1 mutant mice & serotonergic/BD link	Heterozygous haploinsufficiency neuropsychiatric model (tangential)

Limitations and Knowledge Gaps

- **Very small human sample.** The recessive cardiomyopathic form rests on a handful of index cases; prevalence, incidence, sex ratio, penetrance estimates, survival statistics, and the full phenotypic range are therefore poorly quantified.

- **mtDNA-maintenance link is associative.** The connection between ANT1 transport loss and secondary mtDNA deletions/depletion is stated as "probable" [PMID: 27693233]; the precise biochemical pathway (e.g., matrix dNTP/adenine-nucleotide pool imbalance) has not been fully demonstrated for the recessive disease.
- **Compensation biology incomplete.** Why ANT2/ANT3 fail to rescue human heart/muscle, and the quantitative thresholds for compensation, remain to be defined.
- **No therapeutic evidence.** No controlled trials; "mitochondrial cocktail" use is empirical and unproven for MTDPS12B.
- **Cataract mechanism unexplained.** Whether congenital cataracts are a consistent feature or case-specific (and their mechanistic basis) is unresolved; overlap with Sengers-like phenotypes warrants attention in differential diagnosis.

Proposed Follow-up Experiments / Actions

1. **Aggregate a patient registry / genotype–phenotype series** (via GeneMatcher, mitochondrial-disease networks) to quantify prevalence, penetrance, phenotype frequencies, cataract incidence, and natural history.
2. **Directly test the mtDNA-maintenance hypothesis** — measure mitochondrial dNTP/adenine-nucleotide pools and mtDNA deletion burden in ANT1-null human cardiomyocytes/myotubes (iPSC-derived) versus isogenic controls.
3. **iPSC-derived cardiomyocyte and skeletal-myotube models** from recessive patients to characterize bioenergetics (Seahorse), OXPHOS assembly, ROS, and mtDNA copy number, and to screen candidate therapies.
4. **Paralog-rescue experiments:** test whether forced ANT2/ANT3 expression rescues the energy crisis in ANT1-null human striated-muscle models, defining compensation thresholds.
5. **Gene-replacement proof-of-concept** (AAV-SLC25A4) in the *Ant1*−/− mouse heart/muscle, given the recessive LOF mechanism.
6. **Structure-guided variant interpretation:** apply the S1/S2 substrate-site framework [PMID: 42308345] to classify newly found SLC25A4 variants and predict recessive-versus-dominant behavior and severity.
7. **Cardiac natural-history imaging study** to identify prognostic biomarkers (echo/MRI parameters, serial lactate) for the life-limiting cardiomyopathy.

Evidence source types: human clinical case reports (

16155110

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), model-organism studies (mouse: 18439414, 16303948, 20671283, 29892051; yeast: within 16155110), and in-vitro/computational structural work (42308345). Ontology suggestions (MONDO:0014175; HPO HP:0001639, HP:0003198, HP:0003546, HP:0003128, HP:0003200, HP:0000519; GO:0140021, GO:0005743, GO:0000002, GO:0006979; CL:0000746, CL:0000188; UBERON:0000948, UBERON:0001134, UBERON:0000965) are provided to support knowledge-base curation.